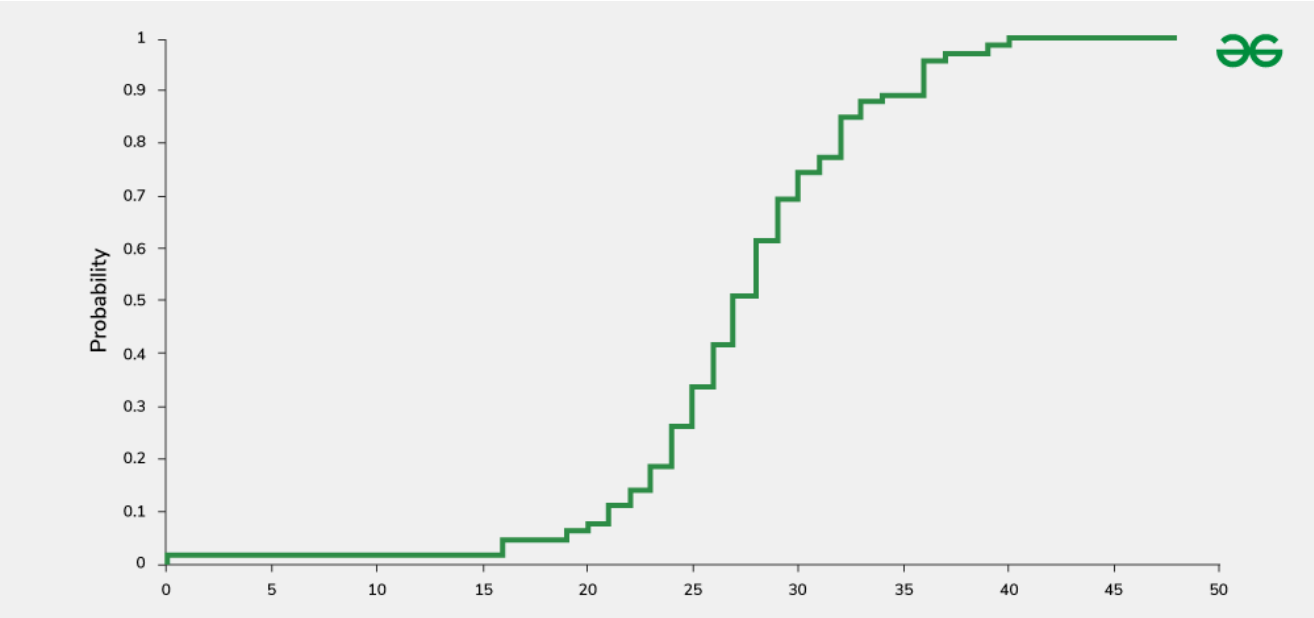
CDF (Cumulative Distribution Function)



The Cumulative Distribution Function (CDF) of a random variable is a mathematical function that provides the probability that the variable will take a value less than or equal to a particular number.

The CDF starts at 0 for the smallest possible value of X and increases to 1 as x approaches the largest possible value of X. It is a non-decreasing function that provides a complete description of the distribution of the random variable.

For example, if you're looking at the CDF for a test score of 80, and it gives you 0.75, this means there's a 75% chance that a random student's score will be 80 or less.

In short, the CDF helps you understand the likelihood of a random value being within a certain range by summing up probabilities as you go along.

$$F(x) = P(X \leq x)$$

Properties of Cumulative Distribution Function

Some of the properties of Cumulative Distribution Function:

- **Monotonicity**

The CDF is a non-decreasing function. This means that for any two values x_1 and x_2 such that $x_1 \leq x_2$ the corresponding CDF values satisfy $F(x_1) \leq F(x_2)$.

- **Limits**

As x approaches negative infinity the CDF approaches 0:

$$\lim_{x \to -\infty} F(x) = 0$$

As x approaches positive infinity the CDF approaches 1:

$$\lim_{x \to +\infty} F(x) = 1$$

- **Continuity**

The CDF of a continuous random variable is a continuous function while the CDF of the discrete random variable has jumps or discontinuities at specific points.

- **Non-Negativity**

The CDF is always non-negative i.e. $F(x) \geq 0$ for the all x.

Type of CDF

CDF of a Discrete Random Variable

For a discrete random variable X with the possible values $x_1, x_2, \ldots, x_n$ and corresponding probabilities $P(X = x_i) = p_i$ the CDF is given by:

$$F(x) = \sum_{x_i \leq x} p_i$$

The CDF of the discrete random variable increases in steps at the points where the variable takes on the specific values.

For example, Consider a random variable X that represents the roll of the fair 6-sided die. The CDF F(x) for this random variable is:

$$F(x) = \begin{cases} 0 & \text{if } x < 1 \\ \frac{1}{6} & \text{if } 1 \leq x < 2 \\ \frac{2}{6} & \text{if } 2 \leq x < 3 \\ \frac{3}{6} & \text{if } 3 \leq x < 4 \\ \frac{4}{6} & \text{if } 4 \leq x < 5 \\ \frac{5}{6} & \text{if } 5 \leq x < 6 \\ 1 & \text{if } x \geq 6 \end{cases}$$

CDF of a Continuous Random Variable

For a continuous random variable X with the probability density function (PDF) f(x) the CDF is the integral of the PDF:

$$F(x) = \int_{-\infty}^{x} f(t)\, dt$$

Since the CDF is the integral of the PDF it is a continuous and differentiable function.

For example: Consider a continuous random variable X that is uniformly distributed between the 0 and 1. The CDF F(x) for this random variable is:

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \leq x \leq 1 \\ 1 & \text{if } x > 1 \end{cases}$$

Difference and relationship between PDF and CDF

Aspect	Cumulative Distribution Function (CDF)	Probability Density Function (PDF)
Definition	Shows the probability that a random variable is less than or equal to a certain value.	Describes the probability of the random variable taking a specific value.
Representation	A running total of probabilities.	The likelihood of the random variable being exactly at a specific value.
Mathematical Expression	F(x) = P(X ≤ x)	f(x) such that f(x) ≥ 0 and $\int_{-\infty}^{\infty} f(x)dx = 1$
Value Range	Always non-decreasing and ranges from 0 to 1.	Can be any non-negative value but does not exceed 1.
Interpretation	CDF gives the cumulative probability up to a certain point.	PDF gives the relative likelihood for a continuous random variable to occur at a specific value.
Visualization	A continuous, non-decreasing curve.	A curve where the area under the curve represents probabilities.
Area Under Curve	Not applicable as it represents cumulative probability.	The total area under the PDF curve equals 1.
Usage	Used to find the probability that a random variable falls within a certain range.	Used to understand the distribution and likelihood of specific outcomes.
Derivative Relationship	The derivative of the CDF gives the PDF.	The integral of the PDF gives the CDF.